



SUPPLY CHAIN DATA ANALYSIS

USING SQL

PROJECT DESCRIPTIONS

This project focuses on analyzing supply chain and order management data using SQL to identify customer behavior, shipment performance, sales trends, and operational efficiency. The analysis was performed using multiple datasets related to orders, customers, products, departments, and shipments.

Dataset Overview

The project uses multiple datasets:

1. orders – Contains order details, shipment details, transaction type, and order status etc.
2. ordered_items – Contains product sales information
3. customer_info – Contains customer details such as city and state
4. product_info – Contains product and department details
5. department – Contains department names

Technology Used

- MySQL Workbench
- PowerPoint for reporting

Data Cleaning

- Checked for null values
- Checked duplicate records
- Verified data types
- Identified **blank spaces** in the **Last_Name** **column(customer_info)**
- Replaced blank variety values with 'Unknown'
- Corrected inconsistent state names (Orders)
- (e.g. "Ammu and Kashmir" → "Jammu and Kashmir")

SQL Concepts Used

- JOINS
- GROUP BY
- ORDER BY
- Aggregate Functions (COUNT, SUM)
- CASE Statements
- CTE (Common Table Expressions)
- HAVING Clause
- Filtering using WHERE conditions

Business Questions Solved

1: Get the number of orders by the Type of Transaction. Please exclude orders shipped from Sangli and Srinagar. Also, exclude the SUSPECTED_FRAUD cases based on the Order Status. Sort the result in the descending order based on the number of orders.

Query

```
SELECT
  Type AS 'Type of Transaction', COUNT(Order_Id) AS Orders
FROM
  orders
WHERE
  Order_City NOT IN ('Sangli' , 'Srinagar')
  AND Order_Status <> 'SUSPECTED_FRAUD'
GROUP BY Type
ORDER BY Orders DESC;
```

Output

Result Grid			Filter Rows:
	Type of Transaction	Orders	
▶	DEBIT	799	
	TRANSFER	532	
	PAYMENT	488	
	CASH	259	

Business Insight: Customer Payment Preferences ,Targeted Market Analysis
Risk-Adjusted Volume

Business Questions Solved

2.Get the list of the Top 3 customers based on the completed orders along with the following details: Customer Id, Customer First Name, Customer City, Customer State, Number of completed orders, Total Sales

Query

```
SELECT
  c.Id AS 'Customer Id',
  c.First_Name AS 'Customer First Name',
  c.City AS 'Customer City',
  c.State AS 'Customer State',
  COUNT(DISTINCT o.Order_Id) AS 'Completed orders',
  ROUND(SUM(oi.Sales), 2) AS 'Total Sales'
FROM
  orders o
JOIN ordered_items oi ON o.Order_Id = oi.Order_Id
JOIN customer_info c ON o.Customer_Id = c.Id
WHERE
  o.Order_Status = 'COMPLETE'
GROUP BY
  c.Id, c.First_Name, c.City, c.State
ORDER BY
  'Completed orders' DESC,
  'Total Sales' DESC
LIMIT 3;
```

Output

	Customer Id	Customer First Name	Customer City	Customer State	Completed orders	Total Sales
▶	5612	Mary	Caguas	PR	3	2188.82
	3465	Mary	Santa Clara	CA	2	2929.74
	7217	Carl	Caguas	PR	2	1239.83

Business Insight:

Identification of VIP/Power Users: It highlights the most loyal, high-frequency customers.

Targeted Customer Retention & Perks: Knowing these specific IDs allows marketing teams to enroll them into VIP loyalty programs, target them with exclusive high-tier rewards, or cross-sell premium products, ensuring they don't churn to competitors.

Business Questions Solved

3.Get the order count by the Shipping Mode and the Department Name. Consider departments with at least 40 closed/completed orders.

Query

```
387 -- Final Output
388
389 • select d.name as Department_Name,o.Shipping_Mode, count(o.Order_Id) as Total_Orders
390 from orders o
391 join ordered_items oi
392 on o.Order_Id=oi.Order_id
393
394 join product_info p
395 on oi.Item_Id = p.Product_Id
396
397 join department d
398 on p.Department_Id=d.Id
399 where o.Order_Status in ('COMPLETE', 'CLOSED')
400 group by d.Name,o.Shipping_Mode
401 having COUNT(o.order_id)>=40 ;
```

Output

Result Grid				Filter Rows:	Search	Export:
	Department_Na...	Shipping_Mode	Total_Orders			
	Fan Shop	Second Class	165			
	Golf	Second Class	75			
	Footwear	Standard Class	86			
	Fan Shop	Standard Class	371			
	Outdoors	Standard Class	73			
	Golf	Standard Class	186			
	Apparel	First Class	84			
	Fan Shop	First Class	119			
	Apparel	Standard Class	300			
	Apparel	Second Class	143			
	Golf	First Class	57			
	Discs Shop	Standard Class	85			

Business Insight:

The analysis of shipping modes and department-wise completed orders showed that some departments consistently received a higher number of successful orders, indicating strong customer demand and better product performance.

These insights can help improve inventory management, shipment planning, and overall supply chain performance.

Business Questions Solved

4.reate a new field as shipment compliance based on Real_Shipping_Days and Scheduled_Shipping_Days.

It should have the following values:

Cancelled shipment - If the Order Status is SUSPECTED_FRAUD or CANCELED

Within schedule - If shipped within the scheduled number of days

On time - If shipped exactly as per schedule

Upto 2 days of delay - If shipped beyond schedule but delay upto 2 days

Beyond 2 days of delay - If shipped beyond schedule with delay beyond 2 days

Output

Result Grid					Filter Rows:		Search	Export:
Order_Id	Shipment_Complian...	Shipping_Mo...	Number_delayed_ord...					
72520	Upto 2 days of delay	Standard Class	466					

```
Query
Limit to 1000 rows

501 -- Final output
502
503 • select Order_Id,shipment_compliance,Shipping_Mode,
504 count(Order_Id) over (partition by Shipping_Mode) as Number_delayed_orders
505 from (
506   select Order_Id,Shipping_Mode,
507   case
508   when Order_Status in ('SUSPECTED_FRAUD','CANCELED')
509   then 'Cancelled Shipment'
510   when Scheduled_Shipping_Days > Real_Shipping_Days
511   then 'Within Schedule'
512   when Real_Shipping_Days = Scheduled_Shipping_Days
513   then 'On time'
514   when Real_Shipping_Days > Scheduled_Shipping_Days +2
515   then 'Beyond 2 days of delay'
516   when Real_Shipping_Days > Scheduled_Shipping_Days
517   and Real_Shipping_Days<=Scheduled_Shipping_Days+2
518   then 'Upto 2 days of delay'
519   Else 'Unknown'
520   end as Shipment_Compliance
521   from orders
522 ) F
523 where shipment_compliance IN ('Upto 2 days of delay','Beyond 2 days of delay')
524 order by Number_delayed_orders desc
525 limit 1;
```

Business Insight:

The shipment compliance analysis helped classify orders based on delivery performance, such as on-time deliveries, delayed shipments, and cancelled shipments.

Business Questions Solved

5. An order is cancelled when the status of the order is either CANCELED or SUSPECTED_FRAUD. Obtain the list of states by the order cancellation % and sort them in the descending order.

Query

```
628 -- final output
629
630 with cancelled as
631 (
632   select Order_State, count(Order_Id) as CANCELED_orders
633   from Orders
634   where order_Status in ('CANCELED', 'SUSPECTED_FRAUD')
635   group by Order_State ),
636
637 total as
638 (
639   select Order_State, count(Order_Id) Total_orders
640   from Orders
641   group by Order_State
642 )
643
644 select c.Order_State,
645        round((c.CANCELED_orders /t.Total_orders)*100,2 )as Cancellation_Percentage
646
647 from cancelled c |
648 join total t
649 on c.Order_state=t.Order_State
650 order by Cancellation_Percentage desc;
651
652
```

Output

Result Grid		Filter Rows:	Search
Order_State	Cancellation_Percenta...		
Manipur	11.11		
Punjab	9.09		
Haryana	9.09		
Jharkhand	8.82		
Puducherry	8.33		
Uttar Pradesh	6.25		
Telangana	5.88		
Andhra Pradesh	5.21		
Jammu and Kashmir	4.55		
Kerala	4.35		
Delhi	4.11		
Karnataka	4.09		
Madhya Pradesh	3.77		
Odisha	3.45		
Chhattisgarh	2.94		
Maharashtra	2.80		
Tamil Nadu	2.55		
Bihar	2.30		
West Bengal	1.64		
Gujarat	1.63		
Rajasthan	0.67		

Business Insight:

The cancellation percentage analysis revealed that some states experienced significantly higher order cancellation rates compared to others.

Key Insights

- Certain transaction types contributed significantly more orders.
- Top customers generated high completed orders and sales revenue.
- Some departments showed consistently high order volumes.
- Standard shipping modes experienced higher shipment delays.
- Cancellation percentages varied across different states.

Challenges faced

- Handling multi-table joins
- Building shipment compliance logic using CASE statements
- Working with aggregate filtering using HAVING
- Managing order status conditions and delay calculations
- Translating business requirements into SQL queries

Conclusion

This project demonstrated how SQL can be used to analyze supply chain operations, shipment performance, customer behavior, and order trends. The analysis improved understanding of SQL querying, data handling, and business problem-solving using relational databases.



THANK YOU

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